

Homework #3

Due date: 23 November 2022

Notes:

- For Question 5, you can use a Python module for arithmetic in $GF(2^8)$.
- You are expected to submit your answer document as well as the Python codes you used.
- Do not submit .ipynb files, **only .py** scripts will be considered. You can work on Colab but please, submit a python file in the end.
- Zip your programs and add a readme.txt document (**if necessary**) to explain the programs and how to use them.
- Name your winzip file as **"cs411_507_hw03_yourname.zip"**

1. (20 pts) You are in a job interview, and you were given the following RSA parameters:

N=

14160376831985083549234691952615806088754482769313863972472612104960426
89722352846577135547157937440663064541821004645700622259511456561023077
13996420730089390645713411633910401928950536677462391987453138366083578
61688352973345896714893493445912599660303023771086463444245877244662588
47549085213970957665207134704078245202042057177305078113263352853462184
26462723475873449307629511159064097300665581109156586029040861523564408
70165207911781076319159782794658036524692436267278201087720505491720755
12900186626664370820817920955813360599193968787528992938952756946761821
88080227736180957494329949550990021380617745580926469350539227687298675
96344560354854024333693063848315568843274016677634235943265653870168403
48375482877866033127357137496266936772314585940742043708721187317938666
95657229502747066847095648259171768460994701758485877123592620114536912
89856193406245598805109425265282471927598683177731795099636499428783962
10326534638486338064556378395711649055109537193994195358884142190120801
82088332319809632527785498703590349866074018950737496211095307392740846
26457126786909062587418357842878093562018076560481599295590002481431272
39879504133444449707805721481828407639971809740271487838163125308653795
89453838465155388268767885312887668258797488906451176135335859947046464
56047742249832303552280560598950442935275040981664802396857172617709467
79751194318388543290206086641657571197777229722339185163614341216688042
00866962243462676619268645778710743511125732382832977176931202124285665
07110945526676138784819788038239839295288788351687768791008155514660231
44343248434020363772531481595174076511654806129561261819805882357684363
95081199323291305539088674790557155643860281233927180922511100770198266
54081948597242333738564418061676295146137851144050784468238754208055080
76613685108066292428166159029301586004194580475473369751095360829939544
1403

C=

69640001348021474903504213656503362496142506108659418645075605309730658
29363013742586720564552761485372116221740627172129123876523265935559212
39814807828333848385588437439080830381566506168724225159624620021697301

54915953964183119114980126908766330985368848309500450339408734729613619
 44443087691593107828862821646570032433642983137001852284845829650930647
 06223012406708943135942308034577714781652449936804193376038948881234784
 23874543403519874324974507113200783002944486585486851600539535628386531
 41801302708015723513455630185668900982623299071334073972206896315094654
 23370485258248363051965708675838699603428210849598364780048917355790608
 65524567162160429678280461454911199062400526031135684121111239955162812
 89156205173721081926194326415619996714900950517971362831818945544753717
 88753487916467074480831971721615478110660543494213591753799041395668218
 40066652595424137210482881658501927325492040686183535597528410306227932
 11811546498498416331840835131156082144938676253315415796019426308810557
 10772521613539736496538502897482744276266996184051595286258970763465218
 55833152050437833779268709820967663507879956765807906582220904644347215
 01852708698323447101795557300364436799578803807748130751797841330640652
 95040609126587134691678763590715741842451613877119115706937778499202286
 16110285482656647558683620463689959080278744496130657075835781539867329
 28925323015151853338913936042198087258685083559185030411159644723900086
 245429733121433845958021799786535702133974611668238336

$$e = 2^{4+1}$$

You are asked to retrieve the plaintext “M” using only these given parameters, the plaintext is a 289-bit number and $M \ll N$ (which means that M is too small than N). Show your work.

From the given hint and the value of e, one can assume that $M^e \ll N$ therefore, there is no need for reduction modulo N, therefore, we can get M by doing the following:

$$C = M^e \bmod N = M^e \Rightarrow \log_2(C) = \log_2(M^e) = e \cdot \log_2(M)$$

$$\log_2(M) = \log_2(C)/e \Rightarrow 2^{\log_2(M)} = 2^{(\log_2(C)/e)} \Rightarrow$$

$$M = 2^{(\log_2(C)/e)}$$

2. (25 pts) Alice encrypts the private factors of the modulus using her public key. In order to increase security, she multiplies them with a random integer k (a process called blinding). Namely, she performs the following operations:

$$c_p = (kp)^e \bmod n \text{ and } c_q = (kq)^e \bmod n,$$

where:

$$n =$$

73512973335060030081468282098336397584185840115543849725566479428982375
 03037718695637830609047406561083968328616312232137592331991101538410373
 74050058977286914202841273524024666932250131821113376668815406723621016
 49097977282533745476584222793770179097317864458142937324148557309714017
 33550710145929124769939738763589008485016472082711182204941268594892277
 94428441209121779400263427064528800044153594488608641089810344852811969
 04768332375221604708731576259326563612846635064979826680730540182601947
 65233835743738656549160897983547411689218105035219148455858391611808212

37530439632455643648718893847945602507596343025039916399868955525098511
 95516884533439960978043849158219089300897385142338631732809146184196710
 17933875694012909890988102141890827802411783985398632856594216125677645
 60272257730862326168958364031351936876932865703535175597367552038394861
 00365686967751078058559507545057056861806132432140658073493899816919764
 70235866303540467163203311155468259406351404248680181614400433190033519
 00977331129699978699948137567053016132949764669941675005253460547841482
 62517489203107663821626294529576908407126832705381935224215695242840900
 51656775595480488299992596031724073766519823034866191167042526714139900
 26284868294201399181512043

$e = 65537$

- a. (15 pts) Explain why this is not secure as anyone who obtains c_p or c_q can factor n .

We know that $c_p = (kp)^e \bmod n$. We can have the following

$$c_p = (kp)^e \bmod pq$$

$(kp)^e = c_p + tpq$ for some integer t . We can say that the left-hand side is divisible by p as $(kp)^e$ is a multiple of p . Then the right-hand side should be divisible by p as well. This leads to the fact that c_p is also divisible by p . As the modulus n is also a multiple of p , then

$$\gcd(c_p, n) = p.$$

- b. (10 pts) Factor n assuming

$$c_p =$$

591585661179913908292670964497762439966067617632980318594933696993454
 439728774695673398116279819100780981035776904828340000286556561821690
 365937795767595578460020372779099849512212679341203336712606691026980
 784826758612415894186878559418781933556513172143678225888892413399407
 357572102187313811291121304928928848821740914239643286098598943861830
 177312939795889893467307582185899931290683200944769576636703683519903
 565708396786917226998954212438208550716658336081133840312436400158765
 503544052687056167181841283624622810770703450396047554977193850903901
 459806510612970231465678503924692665502872536500180880428444652182260
 677241127037607744405052492742074626162505690986835662409717998630436
 892609085891138004339602274790886677062920378513594058185353516270486
 167595057731308599477294190461449818277150803142015005367485865449897
 543408948648598768730426058899226212982674326317239306732825540115470
 519539729209054092010428475270982133241739755670905018636775489391024
 836583829979747436647847343118308007690487133027090643538988141634837
 671142574166059246803646505989258216228280398544311438792645908159200
 659488745926355055587928710053702116250352855728835609501615125001498
 450195370742144016227930442188681846248774022591908284395554

$$c_q =$$

309411729251034251013395318719711706160811892862394890384299153591046
 151499917121473269348428595368341405902181095462992704329677547559777

979434641542836057927150208356368122522229704600457038948359531454800
 468151991865398538394737136726928964793436886497125779317949812565735
 742165451557954880576255424560841686766527870866834024733826068596435
 927601933868356893971916253250737903827232908056159513671751051011444
 556858915046438167636879877207093812917634178160773097343436768913865
 449127177896443942872076146021042760369486915412609972874901144604578
 896888158193243455010603320402839406279239012820373216932302406828263
 523610721896771030364051463816982602691512661678520212734151948259164
 880771727519256923996762844291396810356605766709626469613406423888478
 179326659617558931774201308469110659393420333748745481493594465373505
 620910145316327984388627313690258503451934907257473375836215663070727
 820311136494801887688477512058462477531334799901621825205074964487431
 978969359431368816027376768643419649233036868068405017555516631026700
 926970709260767677416518151837286394000000084320497258684909083576218
 584531666178186869180867507209104151471590432167256468837661447181958
 281610404744566801864938311314315544802024261994226882434417

and decrypt the following ciphertext

$C_m =$

410549325685366799284798358668331291049211237703447489321746464354090
 480951066742200309912524869634695769764148842963115769175527179769353
 501136785040366302701836605559269222653839509896223034315989474752525
 422032311356077141993653194048634554747064603617789881601934904375748
 720866525993835837380575119329831703593942313434602973137707915466599
 022298281760026130881260276637144571320608488322550304670024203670226
 128476075072338279764675774256431888736850559740563151465682499215963
 629918224101757068211315022934434916282345252224165558403932898552995
 551505591198395331625457165173901358070740812251352212374898226752191
 458989544999183050305396079595170576363831967410596235893058231852520
 802407515041647925479992058184613103289709752618729186509661786057001
 647083804585301703801272797236111687348907977621232083642431914363895
 266519463459112358271172160273751026080398345460929751476512586349360
 905588246751625224080678692961596943056437462453579785432258654674275
 520068227719175943507656108554707552928863240951458193510571735005767
 226590398204791730466993980158501690130104327829589781448607958503898
 902133445000532521723138230966914652700465728532524467483612403840506
 027543892612439531633794667013713638856396514285792898762678

If we compute $\gcd(c_p, n)$ we obtain

$p =$

312377467846585687027923057994614509289332354193481125226572992251261
 753379889369817998855225663758734755828842283358167036650051875442945
 385472789138665198242998893844311276835035806651864989738695785784993
 670288252024932475786155494699966463801546183654608560302758813684987
 216212330993921211097100231401546682251759090298693728376935711292148
 333922280753798013939799779091741663032470576337052515973410975933439

CS 411-507 Cryptography

119483816548485519302372777154158392383316288203883135467326768666871
098635775956497868798944897247548051812233964580971719611938914319419
93777300585306389941314658668121625229901206556741141773579396583

$q=n//p$

235333789731478333617321739946551238520237701399595086087711574935423
913839525279116379578830546488991890089232992184137868389193021119467
534712822237798587566472293194305062660919763570440246515901190421715
972296119008828361951189231520291131634233322098413875403137819600376
235411914672045716353805194347258594303510274526766127656817870563571
180796558525595028179492104058570845859373340811398609020127935979980
076964686147927535801620872625350215960557812167440680612435721137465
390463086008941980958806273664675213502042232763980864599798563124431
81628558615659108747503806564476795850687533880993615836041798621

$\phi(n)=$

735129733350600300814682820983363975841858401155438497255664794289823
750303771869563783060904740656108396832861631223213759233199110153841
037374050058977286914202841273524024666932250131821113376668815406723
621016490979772825337454765842227937701790973178644581429373241485573
097140173355071014592912476993973876358900848501647208271118220494126
859489227794428441209121779400263427064528800044153594488608641089810
344852811969047683323752216047087315762593265636128466350649798266807
305401826019476523383574373865654916089798354741168921810503521914845
585839161180821237530439632455643648718893847945602507596343024985145
274111149123033987475374728759618652774879183961059464552255182856664
375863198023998561480768614904809483461544579723685158290584937106521
302148013043031620645856410623781266769146727840405892239256261901445
098072221283778303922238977897326099560205283351504860907240081716161
896839984042193907626622766495931103417564170334727740188640368683668
540940078857228114130688320650088146737985133587448783675861674213304
914400300369494854095569890621790787481993944256608286696315335435161
382130509834376467039861206082473738057825052790408825511858787231970
792781337987930298239020072992909437544430559443789560316840

$d=$

229466544931156909131726604343449911558309005789657832039292837584676
205196518930309082046430692274623639684603359780479482927713416793828
312274912523559185779084296259097623824884157665847758004585409108981
874591976379346211879218031720012465043647678995308485318227694295899
550608000462710953286345278878583450397089196298246745191300569703348
538452647099968302208142030321668506758915813395536110631451957989750
068276759303154072626974061731163544540570524056918687705193141632117
384784246378083086483326685172798611752887757189680526015494614459190
932595506664572074647301578667700720537137974692512481466933018937716
356752548600178620958904753440583468572191334151569242814676965312705
542472701557546707542488724767805783041225833764246567330675741312359
526344536716988843913396777272848823936042763804128711072195330114559

140199634264648256150529361594909745925250156118249766385086445087012
 281972863474848723138346612359541367206510982079398284648961899723237
 367319395046802806518020217213770133173916442281435521425410108326313
 054211925243126115480309950904831990166152709426086572790126536414500
 15097187603463447192020447522512726031781935553861076086731086415374
 925735505610862415293279119172619258798028837977655416564713

Finally, the message is:

"I will never stop loving you. Until the stars fall from the sky, until the world ends. until planets collide and suns wither. Until the moon is extinguished and all seas and rivers dry up. Until I become senile and my memory is eroded. Until my tongue is no longer capable of uttering your name. Until my heart beats for the last time, only then I may stop, I may stop loving you."

3. (20 pts) Consider the combining function given in the following table, that is used to combine the outputs of four **maximum-length** LFSR sequences:

$$F(x_1, x_2, x_3, x_4) = x_1 \oplus x_1x_2 \oplus x_2x_3 \oplus x_2x_3x_4 \oplus x_1x_2x_3x_4.$$

- a. (15 pts) Analyze the function F in terms of three criteria:

- Nonlinearity degree
- Balance
- Correlation

Is this a good combining function? Explain your answer.

The truth table of F is as follows:

x_1	x_2	x_3	x_4	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

Nonlinearity: its nonlinearity degree is 4.

It is not balanced (Check the number of 0s & 1s in inputs and outputs).

F is correlated to x_1 (13/16)

F is not correlated to x_2 (7/16)

F is correlated to x_3 (11/16)

F is not correlated to x_4 (7/16)

It is a bad combining function as it is not balanced and highly correlated to its inputs.

4. **(15 pts)** We challenge you to get the plaintext of a ciphertext C that was calculated using an RSA setting, however, we lost the decryption keys, we only have the following:

$N = 15220196297956469159$

$C = 6092243189299681137$

$e = 2^{16} + 1$

(RSA Encryption: $m^e \bmod N$ | Decryption: $C^d \bmod N$)

Can you retrieve the message using only this information? If yes, show how.

- You are not allowed to use external tools (including online tools).

Just Factorize N into $p \times q$ by brute force. Then retrieve the message.

$p = 3765204637$

$q = 4042329107$

$d = 2563677721698492425$

Message = "Eureka"

5. **(20 pts)** Consider $GF(2^8)$ used in AES with the irreducible polynomial $p(x) = x^8 + x^4 + x^3 + x^2 + 1$. You are expected to query the server using `get_poly()` function which will send you two binary polynomials $a(x)$ and $b(x)$ in $GF(2^8)$. Polynomials are expressed as bit strings of their coefficients. For example, $p(x)$ is expressed as '100011111'. You can use the Python code "`client.py`" given in the assignment package to communicate with the server.

- a. **(10 pts)** You are expected to perform $c(x) = a(x) \times b(x)$ in $GF(2^8)$ and return $c(x)$ as bit string using `check_mult()` function.
- b. **(10 pts)** You are expected to compute the multiplicative inverse of $a(x)$ in $GF(2^8)$ and return $a^{-1}(x)$ using `check_inv()` function.

q5_sol.py